SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

ITA0329- MOBILE COMPUTING

ASSIGNMENT 2 – CO2

CO2 Coverage: Mobile Telecommunication Systems; Multiple Access Techniques (SDMA, TDMA, FDMA, CDMA, OFDMA); GSM Architecture, Services, Radio Interfaces, Localization, Calling, Handover & Security; Advanced GSM Extensions (GPRS, HSCSD, DECT).

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Q1 – Mobile Communication or Mobile Computing?

Problem Context – Field-Service Technology Audit
A utility company equips its field staff with: (a) a ruggedised tablet fetching work orders over cellular data, (b) a laptop plugged into Ethernet at different offices, (c) a fixed wireless router at a remote substation, (d) a push-to-talk handset, and (e) a tablet app that records meter readings offline and synchronises later.

Your Challenge

Investigate whether "wireless" and "mobile" mean the same thing, and whether mobile communication and mobile computing are the same thing.

Before analysis:

- Define at least four of your own classification criteria (e.g., user mobility, wireless link, tolerance of disconnection, location/context awareness, device resource limits).

- Predict how each of the five cases is classified (fixed-wired, nomadic, fixed-wireless, mobile) and whether it is mainly a mobile-communication or a mobile-computing problem, giving reasons.
- Design one additional ambiguous case of your own to test your criteria.

Test Your Reasoning

Build a small simulation of case (e) with link availability of 90%, 60% and 30%. Compare two strategies: send immediately (fail if disconnected) vs store-and-forward queue. Record updates delivered, delayed and lost, and mean delivery delay.

Inquiry

Which characteristics of mobile computing are exposed by your results that a wireless link alone does not solve? Which device categories need application-level changes?

Evidence Expected: Own classification criteria, reasoned predictions, an engineered ambiguous case, simulation counts for both strategies, and a justified conclusion.

Answer:

Mobile communication is the exchange of information between devices through a communication network, particularly when one or more endpoints can move. It focuses on connectivity, radio access, signal transmission, mobility management, handover, and delivery of voice or data over a network.

Mobile computing is the use of portable computing devices and software to process, store, retrieve and exchange information while the user or device is mobile or operates in changing network conditions. It includes the device, application, local storage, user interface, synchronisation, context awareness, security and network connection.

The concepts overlap but are not identical. Mobile communication supplies a communication path. Mobile computing concerns the complete computing experience under mobility, including what happens when the path is slow, unavailable or changes during a task.

Aspect	**Mobile communication**	**Mobile computing**
Main focus	Transmission and connectivity	Useful computation and application behaviour
Typical concerns	Radio link, coverage, handover, channel quality	Offline work, synchronisation, battery, storage, context
Example	Push-to-talk voice call	Offline meter-reading app
Relationship	Enables data exchange	Uses communication but can continue locally without it

Classification Criteria

A reliable classification should not use the presence of Wi-Fi or cellular connectivity as its only test. A device can be wireless but fixed, or mobile but temporarily connected by a cable. The following criteria are therefore used.

Criterion 1 – Physical mobility: Can the device or user move while continuing the task? Criterion 2 – Access medium: Is the current connection wired or wireless? Criterion 3 – Session continuity: Can the activity continue while the device changes location or network attachment? Criterion 4 – Disconnection tolerance: Can the application keep working during an outage and reconcile later? Criterion 5 – Context dependence: Does the application use location, time, identity or device context? These criteria allow the classification to reflect both hardware and application behaviour.

Classification labels are operational descriptions, not mutually exclusive laws. A laptop connected to Ethernet in several offices is nomadic because it is moved between locations, even though each active session uses a fixed wired connection. A rugged tablet using cellular data while carried by a field worker is mobile.

Criterion	Question used in audit	Why it matters
Physical mobility	Does the endpoint move during use?	Separates mobile from fixed equipment
Connection type	Wired or wireless?	Separates fixed-wired and fixed-wireless access
Continuity	Can the session survive movement?	Identifies mobility and handover needs
Outage tolerance	Can work continue offline?	Tests application resilience
Context awareness	Does location/time/device context matter?	Identifies mobile-computing features

Classification of the Five Given Cases

Case (a), ruggedised tablet fetching work orders over cellular data: classify as mobile. The tablet is carried by field staff, uses a wireless wide-area network and must tolerate coverage changes. It is both a mobile-communication case and a mobile-computing case. Cellular access provides connectivity, while local work-order storage, user interaction, security and synchronisation are application concerns.

Case (b), laptop plugged into Ethernet at different offices: classify as nomadic. The laptop changes location between offices, but while in use it is connected through a wired network. Its movement does not require continuous wireless handover. The main issue is nomadic computing and network reconfiguration, rather than mobile radio communication.

Case (c), fixed wireless router at a remote substation: classify as fixed-wireless. The router is installed at a fixed location and uses a radio link as a substitute for a cable. Wireless does not automatically mean mobile. Its dominant issue is fixed wireless communication, link reliability and backhaul availability.

Case (d), push-to-talk handset: classify as mobile. The handset is portable and communication is its primary purpose. The dominant problem is mobile communication, including group-call setup, coverage, delay, channel access and handover. The device may also contain computing functions, but they are secondary to the communication service.

Case (e), tablet app recording meter readings offline and synchronising later: classify as mobile computing if the tablet is carried by the worker; it is also a mobile device even when offline. Its defining requirement is disconnected operation, durable local storage, conflict handling and later synchronisation. Wireless access alone does not solve these application-level problems.

Case	Classification	Main problem	Reason
(a) Cellular tablet	Mobile	Both	Portable device and cellular data application
(b) Ethernet laptop	Nomadic	Mainly computing/network configuration	Moved between sites; wired during use
(c) Substation router	Fixed-wireless	Communication	Fixed endpoint with wireless access
(d) Push-to-talk	Mobile	Mobile communication	Portable voice communication
(e) Offline meter app	Mobile computing	Application resilience	Local operation and later synchronisation

Additional Ambiguous Case

Additional case: a delivery van contains a permanently installed computer connected to a roof-mounted 5G modem. The vehicle moves along a route, but the computer itself is bolted into the vehicle. This case tests whether mobility is assigned to the device, the user, the network endpoint or the service.

The system should be classified as mobile because the vehicle and radio endpoint move through the coverage area, even though the computer is physically fixed inside the van. It is both mobile communication and mobile computing: the modem maintains connectivity and may hand over between cells, while the onboard application stores delivery records, maps and proof-of-delivery data.

The case demonstrates that mobility is not equivalent to a handheld form factor. The important question is whether the endpoint changes its point of attachment or operating location during service. If the van is parked permanently and the modem never moves, the same equipment would instead be a fixed-wireless installation.

Test question	Observation	Classification effect
Does the endpoint change location?	Vehicle travels between delivery points	Supports mobile classification

Does the radio attachment change?	Modem may reselect or hand over cells	Mobile communication requirement
Can application work offline?	Delivery records cached locally	Mobile computing requirement
Does hardware need to be handheld?	No; computer is vehicle-mounted	Handheld form factor is not required

Store-and-Forward Simulation Design

The offline meter-reading application is modelled with 1,000 meter updates. Each update is generated independently. Link availability is represented by the probability that a transmission attempt finds the network available. Three availability levels are tested: 90%, 60% and 30%.

Strategy A – Send immediately: each update is attempted once at generation time. If the link is available, it is delivered; otherwise it is lost. This simple strategy has no retry queue. Strategy B – Store-and-forward: each update is first committed to durable local storage. If the link is unavailable, it remains queued and is transmitted at the next available opportunity. For a compact worked model, assume one transmission opportunity per minute and independent link availability at each opportunity.

For a reproducible implementation, use a fixed seed and store each update's generation time, first successful transmission time and final status. The queue must not silently delete an item after a failed attempt. In a real application, acknowledgements and duplicate protection are needed so that retries do not create duplicate meter records.

Parameter	**Value**	**Reason**
Number of updates	1,000	Large enough for stable proportions
Availability cases	0.90, 0.60, 0.30	Required assignment cases
Immediate strategy	One attempt	Represents fail-if-disconnected
Queue strategy	Retry until delivered	Represents store-and-forward
Opportunity interval	1 minute	Explicit simplifying assumption
Random seed	2026	Reproducibility

Expected Simulation Counts and Interpretation

For the immediate strategy, expected delivered updates are $N \times p$ and expected lost updates are $N \times (1 - p)$. With $N = 1{,}000$, the expected delivered counts are 900, 600 and 300 for link availability of 90%, 60% and 30%, respectively. Expected delayed updates are zero in this simplified one-shot strategy because failed updates are lost rather than retained.

For a store-and-forward queue that retries indefinitely with independent availability p, the probability of eventual delivery approaches 1. The expected number of attempts is $1/p$, so the expected number of transmission opportunities until delivery is 1.11, 1.67 and 3.33.

The following values are analytical expectations, not random-seed simulation output. A Monte Carlo run will vary around them. Record the actual counts and mean delay from the executed program in the final evidence table.

Availability	Immediate delivered (expected)	Immediate lost (expected)	Queued eventual delivery (expected)	Expected queue delay*
90%	900	100	1,000	0.11 min
60%	600	400	1,000	0.67 min
30%	300	700	1,000	2.33 min

What the Simulation Reveals

The experiment exposes four characteristics of mobile computing that a wireless link alone cannot solve. First, intermittent connectivity requires local persistence: data must be saved before attempting transmission. Second, eventual consistency is needed because the server and device may temporarily hold different versions of the record. Third, retry and acknowledgement logic is required to distinguish a temporary failure from successful delivery. Fourth, users need feedback that clearly indicates whether a record is saved locally, queued, synchronised or rejected.

The application should assign a unique identifier to every meter reading. The server should treat repeated uploads with the same identifier as idempotent, meaning a retry does not create a second reading. The client should retain a record until it receives a confirmed acknowledgement. If the server rejects a record because of validation or a conflict, the application should show an actionable error rather than retry forever.

Application-level changes are required for cases (a) and (e), and often for vehicle-mounted mobile systems. The push-to-talk handset primarily needs network-level mobility support, although its user interface and call state must still handle coverage loss. The fixed router needs link redundancy and monitoring, not necessarily offline application logic. The nomadic laptop may need automatic network configuration and secure session restoration.

- Local database or durable queue
- Unique update identifiers and idempotent server writes
- Acknowledgement and retry policy with bounded backoff
- Conflict resolution and validation rules
- Clear user-visible synchronisation state

Q2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic

Problem Context – Hostel-Campus Voice and Telemetry

A cell may use a 2 MHz band per direction. Option 1 is FDMA with 25 kHz channels.

Option 2 is TDMA with 200 kHz carriers, each carrying 8 time slots. Both offer 80 nominal channels.

Your Challenge

Investigate whether equal nominal capacity means equal usable capacity.

- Estimate usable channels after guard bands, guard times, synchronisation and control overheads (state and justify your values).
- Predict which scheme gives better utilisation for (i) continuous voice, (ii) bursty data with about 10% duty cycle, and (iii) a 70/30 voice–data mix, and analyse why.

Design three contrasting traffic datasets, including one edge case of your own (e.g., very short calls or synchronised arrivals). Simulate dedicated-channel allocation under both schemes and record utilisation, blocking and wasted capacity.

Inquiry

Where do the two schemes diverge, and what causes it?

Design Decision

At a campus telemetry site, 90% of devices send a 200-byte report per minute. Is dedicated FDMA/TDMA channel allocation justified? Defend your answer.

Evidence Expected: Capacity calculations with overheads, reasoned predictions, three designed traffic conditions, utilisation/blocking evidence and a justified decision.

Answer:

Frequency Division Multiple Access (FDMA) divides the available spectrum into separate frequency channels. A user is assigned a frequency channel for the duration of the connection. Users can transmit simultaneously on different channels, but guard bands are needed to reduce adjacent-channel interference.

Time Division Multiple Access (TDMA) divides a carrier into recurring time slots. Users share the same carrier frequency but transmit in assigned time intervals. Guard time, timing synchronisation and control information reduce the time available for payload.

The assignment provides 2 MHz per direction, 25 kHz FDMA channels, and 200 kHz TDMA carriers with eight slots each. The nominal arithmetic is 2,000 kHz / 25 kHz = 80 FDMA

channels and 2,000 / 200 × 8 = 80 TDMA time-slot channels. Equal nominal capacity does not guarantee equal usable capacity because guard and control overheads differ.

Scheme	Resource unit	Nominal calculation	Nominal channels
FDMA	25 kHz frequency channel	2,000 / 25	80
TDMA	One of 8 slots on a 200 kHz carrier	(2,000 / 200) × 8	80

Usable Capacity After Overheads

The exact guard-band width, TDMA guard-time duration, synchronisation overhead and control-channel allocation are not specified in the assignment. Therefore, the following values are explicit engineering assumptions for a comparative model, not GSM standard values.

FDMA assumption: reserve 5% of the 2 MHz band for edge and inter-channel guard requirements. Usable spectrum is 1,900 kHz. Dividing by 25 kHz gives 76 usable channels. TDMA assumption: retain all ten 200 kHz carriers, but reserve one of eight slots per carrier for control and timing overhead. Seven payload slots remain per carrier, giving 10 × 7 = 70 usable traffic slots.

These assumptions make the comparison transparent. If a different guard-band or control allocation is used, recalculate the values and state the reason. A proper standards-based report must obtain actual frame, guard and control parameters from a cited technical specification.

Item	FDMA assumption	TDMA assumption
Nominal capacity	80 channels	80 slots
Overhead	5% spectrum guard reserve	1 of 8 slots reserved per carrier
Usable resource	1,900 kHz	10 carriers × 7 slots
Usable traffic capacity	76 channels	70 slots
Capacity retained	95%	87.5%

Traffic Pattern 1: Continuous Voice

Continuous voice produces a relatively steady stream of packets or frames at regular intervals. A dedicated channel is suitable because the resource remains useful for most of the call duration. FDMA provides a continuous frequency resource and TDMA provides recurring slots at predictable intervals.

Prediction: both methods should achieve high utilisation when calls are long and active. FDMA has simple continuous allocation but may waste a complete channel during silence unless voice activity detection or channel sharing is used. TDMA can also waste assigned slots during silence if slots are reserved for the entire call. If speech activity is exploited, a scheduler can reassign idle slots, but that is no longer a purely fixed dedicated allocation.

The simulation should generate calls with long holding times and a high activity factor. Measure offered traffic, accepted calls, blocked calls, allocated resource-time and actual

payload resource-time. Utilisation should be defined as payload resource-time divided by available traffic resource-time, not merely accepted calls divided by capacity.

Measure	Expected FDMA behaviour	Expected TDMA behaviour
Blocking	Rises when all frequency channels are occupied	Rises when all usable slots are occupied
Silence waste	Dedicated channel remains reserved	Reserved slot may remain unused
Delay	Low for admitted continuous stream	Bounded by slot repetition interval
Best fit	Steady continuous traffic	Periodic predictable traffic

Traffic Pattern 2: Bursty Data at 10% Duty Cycle

A 10% duty cycle means that a device is active for approximately one tenth of the observation period. If a dedicated resource is reserved for the entire session, roughly 90% of the reserved resource-time may be idle. This is the central weakness of static dedicated allocation for bursty data.

Prediction: a fixed FDMA channel and a fixed TDMA slot both waste capacity when a user sends only short bursts. TDMA can be more flexible if the scheduler reallocates slots dynamically among active users. FDMA can also use dynamic channel assignment, but a fixed frequency reservation has the same fundamental idle-resource problem.

The dataset should contain many devices that generate short bursts at random times. Record the total time each device is assigned a channel, the time it actually transmits, and the number of requests rejected because all dedicated resources are occupied. To make a fair comparison, use the same traffic arrival trace for FDMA and TDMA.

Metric	Definition
Payload utilisation	Transmitted payload resource-time / total available traffic resource-time
Reservation waste	Reserved but idle resource-time / total reserved resource-time
Blocking probability	Blocked allocation requests / total allocation requests
Mean waiting time	Average time from request arrival to resource allocation

Traffic Pattern 3: 70/30 Voice–Data Mix and Edge Case

For a 70/30 voice–data mix, assume 70% of offered resource demand comes from continuous voice sessions and 30% from bursty data. A hybrid scheduler can reserve a predictable baseline for voice while allowing unused resources to be borrowed by data. If borrowing is permitted, the scheduler must reclaim resources quickly enough to protect voice quality.

Edge case: synchronised arrivals. Imagine that 100 telemetry devices are configured to transmit at exactly the start of each minute. The average offered data rate may be low, but instantaneous demand becomes a sharp burst. A fixed dedicated allocation may block requests even when the average channel utilisation is low. Randomising reporting times or assigning transmission windows spreads the load.

The three datasets should be designed to expose different behaviour: long continuous calls, random short data bursts, and mixed traffic with synchronised telemetry. A good experiment changes one traffic property at a time and retains the same total offered load where possible.

Dataset	Traffic design	Expected observation
A – Continuous voice	Long calls; high activity factor	High utilisation; blocking at high load
B – Bursty data	Short bursts; 10% duty cycle	High reservation waste under fixed allocation
C – Mixed and synchronised	70% voice demand, 30% data; periodic burst arrivals	Peak blocking despite moderate average load

Simulation Method and Evidence Table

Use a discrete-event simulation. Each request has an arrival time, service duration, traffic class and payload activity profile. For FDMA, allocate one of the 76 assumed usable frequency channels. For TDMA, allocate one of the 70 assumed usable traffic slots. A request is blocked if no suitable resource is available at its arrival time. Release the resource when the session ends.

For a first fair model, treat one FDMA channel and one TDMA slot as equivalent dedicated service units. This deliberately isolates the allocation behaviour; it does not claim that the physical data rates are identical. A more advanced model can include different per-resource rates, frame periods and control overhead.

Record the following for each scheme and dataset. Fill the final column from your executed simulation. Do not copy expected qualitative trends as measured values.

Dataset	Scheme	Requests	Accepted	Blocked	Payload utilisation	Wasted capacity
A	FDMA	Run output	Run output	Run output	Run output	Run output
A	TDMA	Run output	Run output	Run output	Run output	Run output
B	FDMA	Run output	Run output	Run output	Run output	Run output
B	TDMA	Run output	Run output	Run output	Run output	Run output
C	FDMA	Run output	Run output	Run output	Run output	Run output

C	TDMA	Run output	Run output	Run output	Run output	Run output

Analysis and Campus Telemetry Decision

The two schemes diverge because FDMA partitions frequency continuously, while TDMA partitions time within a carrier. FDMA needs frequency guard bands and filtering; TDMA needs timing alignment, guard intervals and frame/control overhead. Their behaviour under traffic depends more on the allocation policy and traffic pattern than on the names of the access methods alone.

At the campus telemetry site, 90% of devices send a 200-byte report per minute. A dedicated channel for every device is not justified by this traffic profile. Each device is active for a very small fraction of the minute, so static reservations would leave most assigned resource-time unused. A shared scheduled or contention-based reporting mechanism is more appropriate, with aggregation, randomised reporting times and retry handling.

The decision should also consider peak bursts, reliability and latency. If all sensors must report within a strict short deadline, scheduling capacity must be sized for the peak or reports must be staggered. A dedicated channel might be justified for a small number of critical devices with guaranteed low latency, but not for all 5,000 low-rate devices.

- Avoid one permanently reserved channel per sensor
- Stagger reporting times to reduce synchronised peaks
- Aggregate readings where application semantics permit
- Use acknowledgements and bounded retries for reliable delivery
- Reserve deterministic capacity only for critical low-latency traffic

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem

Problem Context – Crowded Stadium Uplink

Several handsets transmit simultaneously in the same band to one base station, each spreading its data with its own code.

Your Challenge

Design a direct-sequence CDMA uplink simulation. Determine how data bits are spread, how the receiver despreads and decides, and what the processing gain is.

Before executing, predict the bit-error behaviour as the number of users grows for: (i) orthogonal (e.g., Walsh) codes with synchronous users, (ii) random non-orthogonal codes of the same length, and (iii) orthogonal codes where one user transmits at ten times the power. analyse your reasoning.

Implement and test with at least three code lengths (e.g., 8, 16, 64). Record bit errors over a stated number of bits, correlation values and processing gain.

Inquiry

- Why do orthogonal codes still fail when users are unsynchronised or powers are unequal?
- Why is power control essential in CDMA?
- Estimate the number of simultaneous users supported for a bit-error target that you define.

Evidence Expected: A working DS-CDMA simulation, predictions made before execution, bit-error evidence for three code lengths and three conditions, and a reasoned explanation of the results.

Answer:

Code Division Multiple Access (CDMA) allows multiple users to share the same frequency band at the same time by assigning each user a spreading code. In direct-sequence CDMA, each data bit is multiplied by a sequence of chips. The chip rate is higher than the data bit rate, so the transmitted signal occupies a wider bandwidth.

The receiver uses the intended user's code to despread the received composite signal. If the spreading codes are orthogonal and aligned in time, the cross-correlation between different users can be zero over the symbol interval. The receiver sums the chip products and decides whether the result is positive or negative.

Processing gain is approximately the ratio of chip rate to data bit rate, or equivalently the spreading-code length for one bit. In decibels, processing gain is 10 log10(L), where L is the number of chips per bit.

Code length L	Processing gain (linear)	Processing gain (dB)
8	8	9.03 dB
16	16	12.04 dB
64	64	18.06 dB

Spreading and Despreading Calculation

Represent binary data bit 1 as +1 and bit 0 as −1. Let a user's spreading code be a length-L sequence of +1 and −1 chips. To spread one data bit b, multiply every chip in the code by b. For example, if b = +1, the transmitted chip sequence equals the code; if b = −1, every chip is inverted.

The receiver correlates the received chips r[k] with the desired code c[k]. The decision statistic is z = (1/L) × sum(r[k]c[k]). If z is positive, decide bit 1; if z is negative, decide bit 0. Noise and interference perturb z. The magnitude of z indicates the strength of the desired signal relative to interference and noise.

For multiple users, the received signal is the sum of each user's spread sequence, scaled by its received amplitude, plus noise. The desired receiver correlates this sum with the target code. Other users contribute cross-correlation terms. Orthogonality makes those terms zero only under the conditions for which the codes are orthogonal.

Stage	Operation	Purpose
Data mapping	0 → −1; 1 → +1	Convert bits to bipolar form
Spreading	b × code chips	Expand one bit into L chips
Channel	Sum user signals and noise	Model shared uplink
Despreading	Correlate with desired code	Recover desired user's bit
Decision	Sign of correlation	Estimate transmitted bit

Predictions Before Execution

Condition (i): synchronous users with orthogonal Walsh codes. Prediction: in an ideal noiseless channel, users remain separable because the code cross-correlation is zero over the aligned bit interval. Increasing the number of users up to the available orthogonal code set does not itself create multiuser interference. With noise, bit errors still occur, and the number of users may increase total received energy and practical receiver dynamic-range demands.

Condition (ii): random non-orthogonal codes of the same length. Prediction: cross-correlation is generally non-zero. Each additional user adds interference to the desired user's correlation output. Error probability should rise as the number of active users increases, particularly for short codes.

Condition (iii): orthogonal codes but one interferer transmits at ten times the desired user's power. Prediction: ideal synchronous orthogonality can mathematically cancel the interferer, but practical timing errors, channel distortion, imperfect code alignment and receiver dynamic-range limitations can cause residual interference. A tenfold power ratio is 10 dB. The experiment should explicitly separate the ideal synchronous case from a deliberately misaligned or channel-distorted case so the conclusion is not overstated.

Condition	Expected trend as users increase	Main reason
Orthogonal, synchronous, equal power	Low BER until code capacity or noise limit	Zero cross-correlation under ideal alignment
Random non-orthogonal	BER increases	Non-zero cross-correlation interference
Orthogonal, 10× interferer	Ideal case may remain low; practical case degrades with mismatch	Residual interference and near-far sensitivity

Correlation and Orthogonality

For two length-L codes a and b, the normalised correlation is rho = (1/L) × sum(a[k]b[k]). If rho = 0, the codes are orthogonal over the tested interval. If rho is +1 or −1, they are identical or inverted. Intermediate magnitudes indicate partial correlation and potential multiple-access interference.

Walsh codes can be generated from the Hadamard matrix. For a Hadamard matrix H of order L, rows are mutually orthogonal: H H-transpose = L I. Normalising the dot product by L produces 1 for a row with itself and 0 for different rows. The code length must be a power of two for the common recursive construction.

Random codes do not generally have zero cross-correlation. Their average correlation may approach zero across many random trials, but individual pairs can have non-zero values. Increasing code length tends to reduce typical normalised cross-correlation, although it does not guarantee perfect orthogonality.

Correlation result	Meaning	Expected receiver impact
rho = 0	Orthogonal over interval	No ideal multiple-access interference
0 < |rho| < 1	Partially correlated	Residual interference
|rho| = 1	Same or inverted code	Severe inability to separate users

Near-Far Problem and Power Control

The near-far problem occurs when users arrive at the base station with very different received powers. A nearby handset may produce a much stronger signal than a distant handset. In a non-orthogonal or imperfectly orthogonal system, the strong user's interference can overwhelm the weak user's desired signal.

Power control is essential because the base station should receive users at approximately controlled power levels. Open-loop power control estimates path loss from received downlink strength. Closed-loop power control measures uplink quality and commands the handset to increase or decrease transmit power. Practical systems combine mechanisms and use limits to avoid excessive interference or battery drain.

Power control does not make every interference source disappear. It reduces the received-power imbalance so that code separation and receiver detection can operate reliably. The experiment should compare equal-power users, a 10× near-far interferer and a controlled-power case.

- Use received power, not only handset transmit power, when modelling near-far effects
- Keep a defined target received power for each active user
- Apply bounded power-control steps and record convergence
- Measure BER before and after power control

Experimental Design for Three Code Lengths

Test code lengths L = 8, 16 and 64. For each length, run at least three channel conditions: synchronous Walsh codes with equal received power; random non-orthogonal codes with equal power; and a near-far condition with one interferer at ten times the desired user's power. Include a noisy or asynchronous variant to expose practical loss of orthogonality.

For each configuration, transmit a stated number of bits per user, for example 10,000 bits, and calculate BER = total bit errors / total transmitted bits. Record the seed, number of users, code length, signal-to-noise ratio, timing offset, power ratio and correlation matrix. A zero BER in a finite run means no errors were observed in that run; it does not prove the true BER is exactly zero.

To estimate simultaneous-user support, define an explicit target, such as BER no greater than 1%. Increase the active user count one at a time and record the largest count meeting the target in repeated trials. Report the chosen SNR and code construction because the supported-user count is not a universal constant.

Required output	What to record
Bit errors	Per user and total
BER	Errors divided by transmitted bits
Correlation	Pairwise normalised code correlations
Processing gain	L and 10 log10(L)
User capacity	Maximum active users meeting BER target
Reproducibility	Seed and all channel parameters

Interpretation and Conclusion

Orthogonal codes fail to provide perfect separation when users are not synchronised because the receiver's correlation window no longer covers matching chip boundaries. A time shift changes the correlation relationship and can create non-zero cross-correlation. Multipath propagation can also produce delayed copies of a signal, weakening orthogonality.

Unequal received powers make residual correlation more damaging. Even a small cross-correlation coefficient can create substantial interference if the interfering user is much stronger than the desired user. This explains why orthogonality alone is not a complete solution to the near-far problem.

The engineering conclusion is that CDMA capacity depends on code properties, synchronisation, received power control, noise, multipath and receiver design. Code length increases processing gain and can reduce typical random-code correlation, but it does not eliminate the need for power control or timing management.

- State whether the experiment is ideal or includes timing/channel imperfections
- Do not compare BER values from different SNRs as if only code length changed
- Use multiple random trials and confidence-aware interpretation
- Explain why zero observed errors is not proof of zero probability

Q4 – OFDMA – Giving Each User the Subcarriers They Can Use Best

Context – Lecture-Hall Downlink

Eight users share 64 subcarriers. Each user's channel quality varies across subcarriers (frequency-selective fading).

Your Challenge

- Before designing, determine how subcarrier spacing relates to symbol duration, why a cyclic prefix is needed, and verify orthogonality of at least three subcarriers numerically (FFT or inner product).
- Design three allocation approaches: fixed contiguous blocks, interleaved round-robin, and channel-aware best-subcarrier.
- Predict total throughput and fairness (e.g., Jain's index) of each, with reasons.

Implement using generated fading gains with fixed seeds over at least 1,000 channel realisations. Record sum throughput, per-user throughput and fairness.

Inquiry

Why can maximising total throughput starve some users? How does the number of users change the benefit of channel-aware allocation?

Design Decision

Select an allocation for a lecture hall in which every student must receive at least a minimum rate that you define and justify.

Evidence Expected: Numerical orthogonality verification, predictions, three allocation schemes implemented, throughput and fairness results, and a justified allocation choice.

Answer:

For an OFDM system with N equally spaced subcarriers, the useful symbol duration T is related to subcarrier spacing Δf by $\Delta f = 1/T$. The subcarriers are orthogonal over the useful symbol interval because the integral of the product of two different subcarrier complex exponentials over T is zero when their frequency difference is an integer multiple of 1/T.

For example, if $\Delta f = 15$ kHz, then $T = 1/15{,}000$ seconds = 66.67 microseconds. This is an illustrative parameter choice for demonstrating the relationship, not a claim that the assignment mandates a 15 kHz numerology.

A cyclic prefix is a copy of the last part of the OFDM symbol placed before the symbol. If the channel's maximum delay spread is shorter than the cyclic-prefix duration, the prefix can absorb delayed multipath components and preserve circular convolution over the useful symbol

interval. The receiver removes the prefix before applying the FFT. The prefix consumes time and reduces spectral efficiency.

Illustrative parameter	Value	Calculation
Subcarrier spacing	15 kHz	Assumed
Useful symbol duration	66.67 μs	1 / 15,000
Cyclic prefix	4.7 μs	Illustrative assumption
Total symbol duration	71.37 μs	66.67 + 4.7
Useful-time fraction	93.4%	66.67 / 71.37

Numerical Verification of Orthogonality

Consider subcarriers at frequencies f_k = kΔf and f_m = mΔf. Their normalised inner product over one useful symbol is (1/T)∫ from 0 to T of exp(j2π(f_k − f_m)t)dt. If k = m, the value is 1. If k and m are different integers and Δf = 1/T, the integral is zero.

For a discrete implementation with N samples, generate x_k[n] = exp(j2πkn/N). Compute the normalised inner product C(k,m) = (1/N)Σ x_k[n] × conjugate(x_m[n]). For distinct subcarriers, the magnitude should be close to zero up to floating-point precision. Use at least three subcarriers, for example k = 0, 1 and 2, and report the 3 × 3 correlation matrix.

The diagonal entries should be approximately 1 and off-diagonal entries approximately 0. A small non-zero value such as 10^-15 is numerical round-off, not meaningful interference. If the cyclic prefix is included, orthogonality is assessed over the useful symbol after prefix removal.

Subcarrier pair	Expected normalised inner product
(0,0), (1,1), (2,2)	1
(0,1), (0,2), (1,2)	0
(1,0), (2,0), (2,1)	0

Allocation Strategy 1: Fixed Contiguous Blocks

Divide the 64 subcarriers into eight contiguous blocks of eight subcarriers each. User 1 receives subcarriers 0–7, user 2 receives 8–15, and so on. The allocation is simple, predictable and has low scheduling complexity. It can be useful when users need stable resource assignments.

Its weakness is that a user may receive a block containing poor channel gains even when good subcarriers exist elsewhere in the band. A contiguous block may also experience a deep fade over several neighbouring subcarriers. The method does not adapt to instantaneous frequency-selective conditions.

The experiment should calculate each user's throughput from the sum of the spectral efficiencies on the assigned subcarriers. If the Shannon-style proxy is used, define rate per subcarrier as B × log2(1 + SNR), and state that this is an idealised upper-bound model that omits coding, modulation constraints and protocol overhead.

Property	Assessment
Complexity	Low
Predictability	High

Frequency diversity	Limited
Fairness	Usually balanced in resource count, not necessarily in rate
Main limitation	May allocate weak subcarriers to a user

Allocation Strategy 2: Interleaved Round-Robin

Assign subcarriers cyclically: user 1 receives 0, 8, 16, 24, 32, 40, 48 and 56; user 2 receives 1, 9, 17, 25, 33, 41, 49 and 57; continue for all eight users. Each user receives eight subcarriers spread across the band.

Interleaving can provide frequency diversity because a narrowband fade is less likely to affect all of one user's allocated subcarriers. The scheme is fair in the number of assigned subcarriers and does not require instantaneous channel-state information for allocation.

However, it does not exploit which subcarriers are currently best for each user. If channel quality varies strongly by user and frequency, channel-aware scheduling can achieve a higher sum throughput. Interleaving is a useful baseline because it is simple and relatively robust.

Property	**Assessment**
Complexity	Low
Frequency diversity	High relative to contiguous allocation
Channel feedback requirement	Low
Fairness in subcarrier count	Equal
Main limitation	Ignores instantaneous channel quality

Allocation Strategy 3: Channel-Aware Best-Subcarrier

For each subcarrier, estimate the channel quality of all eight users and assign that subcarrier to the user with the highest achievable rate or utility. This greedy strategy attempts to maximise the sum throughput in each scheduling interval.

The method exploits multiuser diversity: different users may have favourable channel conditions on different subcarriers. If channel estimates are accurate and sufficiently fresh, the scheduler can choose the user who can use each subcarrier most efficiently.

The disadvantage is that a user with persistently poor channel gains may receive very few subcarriers. A pure best-channel scheduler can starve weak users. The algorithm also requires channel-state feedback, scheduling computation and signalling. A practical scheduler should add a minimum-rate constraint or proportional-fair utility.

Property	**Assessment**
Sum throughput	Potentially highest
Fairness	Can be low without constraints
Feedback	Requires per-user channel information
Complexity	Higher than fixed/interleaved
Main limitation	Strong users may dominate allocations

For each user i, calculate throughput R_i as the sum of the rates of the subcarriers assigned to that user. Aggregate throughput is ΣR_i. Jain's fairness index is J = (ΣR_i)^2 / (U × ΣR_i^2), where U is the number of users. The index lies between 1/U and 1 for non-negative rates. A value of 1 means equal throughput across users.

Use at least 1,000 channel realisations and a fixed random seed. For each realisation, generate frequency-selective fading gains for eight users and 64 subcarriers. Apply all three allocation schemes to the same channel-gain matrix. This paired design ensures the schemes are compared under equivalent conditions.

Report average sum throughput, average per-user throughput, Jain's index, and preferably the distribution across realisations. Do not report only a single favourable channel snapshot. Include a graph of per-user average throughput so that starvation is visible.

Metric	**Formula or reporting method**
Per-user throughput	Sum of allocated subcarrier rates
Sum throughput	Σ R_i
Jain's index	(Σ R_i)^2 / (U Σ R_i^2)
Channel trials	At least 1,000
Reproducibility	Fixed seed and identical gains per strategy

Why Throughput Can Starve Users and Final Decision

Maximising total throughput rewards allocations that produce the largest immediate rate. If a particular user has strong channel conditions across many subcarriers, assigning resources to that user can increase the total rate more than serving a weak user. The greedy algorithm therefore has no inherent reason to provide a minimum rate to every student.

The benefit of channel-aware allocation generally increases when there are more users because the scheduler has more opportunities to find a user with a strong channel on each subcarrier. This multiuser-diversity gain is not unlimited. It depends on channel independence, feedback quality, user mobility, scheduling interval and the cost of signalling.

For the lecture hall, choose a constrained proportional-fair or minimum-rate-aware scheduler. First reserve enough resources to meet each student's defined minimum rate where feasible; allocate remaining subcarriers using channel-aware selection. Define the minimum rate from the service requirement, for example 256 kbit/s per active student in the model, and verify that the total required rate is feasible under the assumed bandwidth and spectral efficiency. If infeasible, report the admission-control or load-shedding policy rather than claiming all users can be guaranteed service.

- Define a minimum per-user rate before running the experiment
- Allocate baseline resources to users below the minimum
- Use channel-aware scheduling for remaining resources
- Report both sum throughput and minimum-user throughput
- State when the total demand exceeds system capacity

Q5 – SDMA – Reusing the Same Channel in Different Directions 6 Marks Problem
Context – Highway Cell With Directional Antennas

Vehicles along a highway use the same frequency and time slot. The base station can serve them with an omni antenna, a 120° sector antenna, or an N-element uniform linear array steered toward each user.

Your Challenge

- Derive the array factor of an N-element uniform linear array and predict how main-lobe width and side-lobe level change with N.
- Define an SIR threshold and predict the minimum angular separation at which two users can share the same resource.
- Compute patterns for N = 2, 4, 8, 16 and evaluate the SIR at user B when the beam is steered at user A, for separations of 5°–60°. Record beamwidth, side-lobe level and the minimum separation meeting your threshold.

Inquiry

What limits SDMA in practice (angular spread, clustered users)? Why can two users in nearly the same direction not be separated?

Design Decision

At a toll plaza most vehicles arrive in the same direction. Would you rely on SDMA there? Defend your decision.

Evidence Expected: Array-pattern derivation, predictions, computed patterns and SIR results across angular separations, and a reasoned explanation of SDMA limits.

Answer:

Space Division Multiple Access (SDMA) separates users by their spatial direction or spatial channel. Users may share the same frequency and time resource when the base station can distinguish their signals using directional antennas, beamforming or multiple antenna arrays.

The assignment compares an omni antenna, a 120-degree sector antenna and an N-element uniform linear array (ULA). An omni antenna radiates broadly, a sector antenna covers a fixed angular region, and a steered array forms a narrower directional beam toward a selected user.

The central engineering question is whether the beam directed at user A provides sufficient attenuation toward user B. The answer depends on angular separation, array size, element spacing, wavelength, beamforming weights, side lobes and propagation conditions.

Antenna	Spatial selectivity	Typical use

Omnidirectional	Low	Broad coverage
120° sector	Moderate	Sectorised cellular coverage
ULA beamforming	Higher; depends on N and geometry	Separate users by direction

Derivation of the ULA Array Factor

Consider an N-element uniform linear array with element spacing d. Let θ be the observation angle measured from broadside, and let the phase difference between adjacent elements be $\psi = (2\pi d/\lambda)\sin\theta + \beta$, where λ is wavelength and β is the progressive phase shift applied by the beamformer.

For equal-amplitude excitation, the array factor is $AF(\theta) = \Sigma$ from n=0 to N−1 of $\exp(jn\psi)$. This is a finite geometric series. Its magnitude is $|AF(\theta)| = |\sin(N\psi/2) / \sin(\psi/2)|$. The normalised array factor divides this magnitude by N so the peak equals one.

To steer the main beam toward θ0, choose $\beta = -(2\pi d/\lambda)\sin\theta_0$. The phase term then becomes $\psi = (2\pi d/\lambda)(\sin\theta - \sin\theta_0)$. For half-wavelength spacing $d = \lambda/2$, $\psi = \pi(\sin\theta - \sin\theta_0)$. This is the expression used in the simulation.

Symbol	**Meaning**
N	Number of antenna elements
d	Spacing between adjacent elements
λ	Wavelength
θ	Observation angle
θ0	Desired steering direction
β	Progressive phase shift

Main-Lobe Width and Side Lobes

Increasing N while keeping element spacing fixed narrows the main lobe. A narrower beam provides greater angular discrimination and can reduce interference toward users located away from the target direction. The approximate beamwidth is inversely proportional to the array aperture, which grows with N.

For a uniformly excited broadside array, the first nulls occur approximately where the phase difference reaches $\pm 2\pi/N$. The approximate first-null beamwidth in sine-angle space is therefore proportional to 2/N. Exact angular beamwidth depends on steering direction and the sine transformation.

For a uniformly weighted array, the side-lobe level is governed by the excitation window. Increasing N primarily narrows the main lobe; it does not automatically eliminate side lobes. Amplitude tapering can reduce side lobes but broadens the main lobe and may reduce peak gain. The simulation should report the side-lobe level relative to the main-lobe peak using a clearly stated exclusion region around the main lobe.

SIR Threshold and Angular Separation

Define the signal-to-interference ratio (SIR) at user B when the beam is steered toward user A as SIR_B = P_B × G(θ_B) / [P_A × G(θ_B relative to beam A) + I_other]. For a simplified two-user test with equal transmit powers and no other interference, evaluate the beam gain toward B relative to the gain toward A.

For a practical design target, define an SIR threshold of 10 dB as an explicit modelling assumption. This means the desired received power should be at least ten times the interference power. The assignment does not prescribe a threshold; it must be selected and justified based on the service quality being modelled.

The minimum angular separation is the smallest tested separation for which the calculated SIR meets or exceeds 10 dB. Because the array pattern contains side lobes and nulls, SIR need not increase monotonically with angle at every point. Evaluate the full 5°–60° range and report the smallest passing separation for each N.

Model input	**Assumption**
Element spacing	λ/2
Element weights	Equal amplitude
User powers	Equal received power unless stated
SIR threshold	10 dB
Angular separation sweep	5° to 60°
Array sizes	N = 2, 4, 8, 16

Pattern Computation and Measurement

For each N in {2, 4, 8, 16}, calculate the normalised array factor across an angular grid from −90° to +90°. Steer the beam to user A, which may be defined at broadside (0°). Evaluate the gain at user B for separations from 5° to 60°.

Measure half-power beamwidth (HPBW) as the angular width between the two points where the normalised power gain falls to 0.5, equivalent to −3 dB. Measure peak side-lobe level as the largest local maximum outside the main-lobe region, expressed as 20 log10 of the normalised field magnitude. Record SIR at every separation.

The exact HPBW and side-lobe values should be computed from the numerical array pattern rather than copied from a generic table. Use sufficient angular resolution, for example 0.05 degrees, to avoid missing narrow peaks in the 16-element pattern.

N	**HPBW**	**Peak side-lobe level**	**Minimum separation meeting 10 dB SIR**
2	Compute from code	Compute from code	Compute from code
4	Compute from code	Compute from code	Compute from code

8	Compute from code	Compute from code	Compute from code
16	Compute from code	Compute from code	Compute from code

Practical Limits of SDMA

Angular spread limits SDMA because a real user signal may arrive over a range of directions due to reflections and scattering rather than from one exact angle. A beamformer designed for one direction may not suppress all multipath components. If two users are physically close or appear in nearly the same direction, their spatial signatures can be highly correlated.

Clustered users are difficult to separate with a simple single-beam ULA because the array may not have enough spatial degrees of freedom to form distinct beams and nulls. The number of independently separable spatial streams is constrained by the number of antenna elements, channel rank, signal-to-noise ratio and propagation environment.

At a toll plaza, most vehicles arrive in the same direction and may be tightly clustered in angle. SDMA alone is therefore not a reliable primary separation mechanism. A better design combines spatial sectors with time scheduling, frequency allocation or vehicle-specific short scheduling windows. SDMA can still help separate lanes or different approach directions where angular separation is sufficient.

- Use SDMA where users have distinguishable spatial signatures
- Combine SDMA with TDMA or OFDMA for angularly clustered traffic
- Avoid assuming that more antenna elements solve all propagation problems
- Test realistic multipath and unequal-power cases in an extended model

Q6 – Choosing Multiple-Access Techniques for Real Deployments 7 Marks

Problem Context – Four Deployments

Choose access methods for: (1) wide-area cellular voice, (2) a jam-resistant tactical link, (3) a high-rate stadium downlink, (4) a network of 5,000 low-rate sensors sending small reports.

Your Challenge

- Define at least five weighted criteria and justify the weights yourself.
- Rate each of SDMA, TDMA, FDMA, CDMA and OFDMA for each deployment using your own Q2–Q5 evidence wherever possible.
- Choose a primary technique and, where useful, a hybrid for each deployment.
- Perform a sensitivity check by changing weights by ±20%: does any choice change?

Inquiry

Why do real systems combine techniques (for example FDMA with TDMA)? Which of your choices is least robust, and why?

Evidence Expected: Independently justified criteria and weights, ratings traceable to your own Q2–Q5 evidence, sensitivity analysis and defended choices.

Answer:

The four deployments have different operating objectives. Wide-area cellular voice prioritises coverage, predictable delay and continuity. A tactical link may prioritise resistance to interference and interception, although no multiple-access method alone guarantees security or jamming resistance. A high-rate stadium downlink needs high spectral efficiency and scheduling flexibility. A 5,000-sensor network needs scalable low-overhead access for small, infrequent reports.

The five techniques are SDMA, TDMA, FDMA, CDMA and OFDMA. A weighted decision matrix makes the selection traceable. Scores should be assigned consistently and justified using the Q2–Q5 findings. The scores below are an initial engineering judgement for the stated deployment, not universal rankings.

Criterion	Weight	Reason
Spectral efficiency	25%	How much useful data is carried per unit spectrum
Scalability	20%	Ability to support many users
Latency and predictability	20%	Importance of timely delivery

Interference robustness	20%	Ability to operate with interference and imperfect channels
Implementation complexity	15%	Cost of scheduling, synchronisation and hardware

Rating Scale and Weighted Method

Use a five-point rating scale: 1 = poor, 2 = below average, 3 = acceptable, 4 = good, 5 = excellent. For each technique and deployment, multiply the rating by the criterion weight and add the weighted scores. The maximum score is 5.0.

The weights sum to 100%. The selected weights deliberately emphasise spectral efficiency, scalability, latency and interference robustness. If the deployment has a different priority—for example, a tactical link may place greater emphasis on interference robustness—the weights should be adjusted and the ranking recalculated.

Do not choose a method merely because it has the highest weighted score. The final design should also account for whether the method is technically suitable, whether it can be implemented with the available equipment, and whether a hybrid provides a better engineering compromise.

Rating	**Interpretation**
1	Poor fit
2	Weak fit
3	Acceptable / conditional fit
4	Good fit
5	Very good fit

Deployment 1: Wide-Area Cellular Voice

Recommended primary approach: a hybrid of FDMA and TDMA for a traditional narrowband cellular architecture, with sectorisation or SDMA where spatial reuse is practical. FDMA separates carriers and TDMA divides each carrier into recurring time slots. This combination allows a cell to allocate multiple voice channels over a set of radio carriers.

The hybrid is useful because frequency separation and time scheduling address different dimensions of resource sharing. A fixed frequency channel can be divided into multiple time slots, increasing the number of logical traffic channels while retaining a manageable carrier plan.

The choice is least appropriate for a modern high-throughput broadband voice/data network if interpreted as the only possible architecture. OFDMA-based systems can provide flexible resource-block scheduling and integrate voice as packet traffic. The answer should therefore identify the assumed legacy/narrowband voice context and avoid claiming that FDMA/TDMA is universally best.

Technique	**Fit**	**Reason**

FDMA	Good	Carrier separation and predictable channel planning
TDMA	Good	Recurring time slots for voice streams
CDMA	Good/conditional	Multiple users share spectrum; requires power control
OFDMA	Very good for broadband	Flexible packet scheduling
SDMA	Supporting hybrid	Spatial reuse and sectorisation

Deployment 2: Jam-Resistant Tactical Link

A direct-sequence spread-spectrum CDMA-like approach may be selected as a candidate because spreading can provide processing gain against certain narrowband interference and can make the signal less conspicuous. However, spreading is not equivalent to encryption, anti-jam security or guaranteed resilience against a capable wideband jammer.

The tactical design should combine spreading with frequency agility, robust coding, authentication, encryption, power control and adaptive rate selection where permitted. The access method is only one part of a secure radio system. The exact choice depends on threat model, spectrum rules, synchronisation capability and hardware constraints.

The assignment asks for a primary technique and useful hybrid. Select CDMA/spread-spectrum as the primary multiple-access concept, supported by frequency hopping or channel agility as a separate anti-interference mechanism. State that the detailed anti-jam performance requires a threat-specific link simulation beyond a simple CDMA BER model.

Technique	**Fit**	**Reason**
CDMA	Good candidate	Processing gain against some narrowband interference
FDMA	Conditional	Can avoid occupied bands but fixed channels may be jammed
TDMA	Conditional	Time scheduling does not itself resist jamming
OFDMA	Conditional	Flexible subcarriers but wideband jamming can affect many
SDMA	Supporting	Directional antennas can reduce off-axis interference

Deployment 3: High-Rate Stadium Downlink

Recommended primary technique: OFDMA, supported by SDMA/beamforming. OFDMA divides the bandwidth into subcarriers and assigns resource blocks to users based on demand

and channel quality. This is suitable for a dense stadium where users have different channel conditions and traffic demands.

SDMA can increase spatial reuse by directing beams toward different user groups. Combining OFDMA and SDMA allows separation in both frequency-time resources and spatial direction. However, clustered spectators, blockage and rapidly changing channels can limit beamforming gains.

The scheduler should use proportional fairness or minimum-rate constraints rather than pure best-channel allocation. This avoids a design in which a small number of users receive nearly all the high-rate resources while weak users are starved.

Technique	Fit	Reason
OFDMA	Very good	Flexible high-rate resource-block scheduling
SDMA	Very good supporting method	Spatial reuse and beamforming
TDMA	Moderate	Can schedule users but less flexible in frequency
FDMA	Moderate/low	Rigid channel allocation for bursty broadband traffic
CDMA	Moderate/low	Interference-limited under heavy high-rate load

Deployment 4: 5,000 Low-Rate Sensors

For 5,000 sensors sending small, infrequent reports, static dedicated FDMA or TDMA allocation to every device is inefficient. Most devices would hold a resource while idle. A scheduled or contention-based packet-access design with aggregation, randomised reporting and low-overhead signalling is more suitable.

OFDMA can provide flexible scheduled access when the network controls the devices and has suitable synchronisation. A lightweight contention-based access mechanism may be preferable for sporadic uplink reports. The exact radio technology depends on range, battery life, coverage, payload size, reporting deadlines and the ability to synchronise devices.

Choose an OFDMA-based shared resource pool with grant-based scheduling for devices that need reliability, combined with random-access or grouped reporting for sporadic low-rate devices. Do not claim that OFDMA alone solves access congestion; random access collisions, control-channel capacity and device synchronisation must be modelled.

Requirement	Design response
5,000 devices	Shared pool and scalable scheduling
Small payload	Minimise control overhead
Infrequent reports	Avoid permanent dedicated reservations
Battery operation	Reduce unnecessary wake-ups and retransmissions

Weighted Decision Matrix and Sensitivity

A complete matrix should be prepared for each deployment, with five techniques rated against the five criteria. To make the result defensible, write a one-sentence justification for every rating. A high score for CDMA in the tactical case, for example, should refer to the spreading and interference experiments in Q3 rather than a generic claim.

Perform sensitivity analysis by changing each weight by +20% and −20%, then renormalising all weights so the total remains 100%. Recalculate the weighted scores. If the winning technique changes under small weight adjustments, describe the decision as sensitive and explain which criterion drives the change. If the winner remains the same, the choice is more robust to the tested weighting changes.

Because the assignment requests your own ratings and weights, the following template should be filled with your final judgement and supported by the Q2–Q5 evidence. The report should show both the baseline and the changed-weight result.

Deployment	Primary choice	Useful hybrid	Key risk
Wide-area cellular voice	FDMA/TDMA (assumed narrowband system)	SDMA/sectorisation	Legacy assumptions may not fit broadband systems
Tactical link	CDMA/spread-spectrum candidate	Frequency agility + SDMA	Not a guarantee against all jamming
Stadium downlink	OFDMA	SDMA/beamforming	Fairness and feedback overhead
5,000 sensors	Shared scheduled access / OFDMA candidate	Random access and aggregation	Control congestion and synchronisation

Why Real Systems Combine Techniques

Real systems combine access techniques because no single dimension is sufficient for all traffic and deployment conditions. FDMA separates users in frequency, TDMA separates them in time, CDMA separates them by code, OFDMA separates them across orthogonal subcarriers and time scheduling, and SDMA separates users spatially.

A cellular system may use frequency channels and time slots together because the two methods partition different resources. A broadband system may combine OFDMA with SDMA because frequency scheduling and beamforming can be applied simultaneously. The combination can improve capacity, but it adds coordination, feedback, synchronisation and implementation complexity.

The least robust choice in the proposed set is the pure greedy best-subcarrier OFDMA scheduler for a fairness-critical classroom, because its throughput advantage can come at the expense of weak users. It becomes more robust when minimum-rate guarantees or proportional-fair scheduling are introduced. A pure SDMA design is also fragile when users are angularly clustered.

- Explain the role of each access dimension separately
- Use hybrid methods only when the added complexity provides a measurable benefit
- Report sensitivity rather than presenting one score as absolute truth
- Identify the least robust decision and the conditions under which it fails

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate **13 Marks**

Review the evidence you generated while working on Q1–Q6.

Select **ONE** solution that demonstrates a significant limitation as its problem size or operating requirement increases. Examples of limitations you may find in this variant:

- Dedicated FDMA/TDMA channels sit idle under bursty traffic.
- A CDMA link collapses under power imbalance or loss of synchronisation.
- Best-subcarrier OFDMA allocation starves weak users.
- SDMA fails when users are angularly clustered.

You are not permitted simply to replace the solution with a more advanced technology or standard and state that it is better. Your task is to understand the problem first and then develop an appropriate redesign.

A. Diagnose the Bottleneck **2 Marks**

- Where most capacity, signalling or computational effort is being spent.
- Which counted quantity grows significantly as load or size increases.
- Whether resources are idle, wasted or repeated unnecessarily.
- Which characteristic of the original design causes the limitation.

Evidence Expected: A precise diagnosis supported by evidence from your original implementation and analysis.

B. Define the Design Objective **2 Marks**

- State what your redesign must achieve before proposing it (e.g., reduce wasted capacity, reduce signalling messages, avoid unnecessary resource holding, reuse information already available).
- Specify a measurable objective wherever possible.

Evidence Expected: A clearly stated, measurable objective derived directly from the

- What you have changed and what property or information the new design exploits.
- Which operations or resource usages are avoided or reduced.
- Whether additional state, preprocessing or complexity is introduced, and any new trade-offs.
- Provide pseudocode, a flowchart or a message-sequence design.
- Marks are based on the quality of reasoning, not merely on naming a known technique.

Evidence Expected: A coherent redesign directly addressing the bottleneck, with a design representation and technical justification.

C. Predict Before Implementing **1 Mark**

- Predict how the redesigned solution should differ from the original and in which measure you expect improvement.

Evidence Expected: A clear, testable prediction made before experimental validation.

D. Implement and Validate the Redesign

- Implement a prototype or simulation and compare Original → Redesigned under the same or equivalent test conditions.
- Use counted measures (e.g., slots used, messages, blocking, bit errors, throughput, fairness); run-time is supporting evidence only.

Evidence Expected: Comparable experimental evidence for the original and redesigned solutions.

E. Defend the Final Design

- Was the prediction supported, and what was the magnitude of improvement?
- What new costs or trade-offs appeared, and when might the original still be preferable?
- A redesign that does not outperform the original is not automatically unsuccessful if you correctly analyse why the hypothesis failed.

Evidence Expected: A critical, evidence-based evaluation and a defensible final engineering conclusion.

Answer:

For the redesign challenge, select the OFDMA best-subcarrier allocation from Q4 as the original solution. It demonstrates a significant limitation as the number of users increases or as some users experience consistently poor channel quality: a pure greedy scheduler can starve weak users even while aggregate throughput remains high.

This choice is suitable because it is supported by the Q4 experiment. The original method assigns each subcarrier to the user with the highest instantaneous rate. It optimises the sum rate locally but has no memory of how much service each user has received in previous scheduling intervals.

The redesign will retain OFDMA and the same 64 subcarriers. It will not replace the system with a different advanced standard. Instead, it changes the scheduling objective by adding a minimum-service rule and a proportional-fair selection step.

Design element	**Original approach**
Resource pool	64 OFDMA subcarriers
Users	8
Allocation	Best instantaneous channel per subcarrier
Objective	Maximise immediate sum throughput
Observed limitation	Weak users may receive very few resources
Redesign target	Protect minimum per-user service while retaining channel awareness

Q7A – Diagnose the Bottleneck (2 Marks)

The main bottleneck is not the number of subcarriers alone; it is the greedy allocation rule. For each subcarrier, the scheduler compares the instantaneous rates of all users and assigns the resource to the user with the highest rate. If a user has poor channel quality over many scheduling intervals, it repeatedly loses the comparison.

The counted quantity that grows with load is the number of user-subcarrier comparisons. With U users and S subcarriers, a direct greedy assignment requires approximately $U \times S$ rate comparisons per scheduling interval. For $U = 8$ and $S = 64$, this is 512 comparisons per interval. This computation is manageable at the stated size, but it scales linearly with the user count for a fixed number of subcarriers.

The more serious resource limitation is service imbalance: weak users receive too few subcarriers, while strong users repeatedly consume the available resources. The wasted resource is not necessarily an idle subcarrier; it is an allocation policy that fails the service objective for some users. The evidence should be the per-user throughput distribution and Jain's fairness index from Q4.

Diagnostic item	**Finding**	**Evidence to report**
Capacity/signalling effort	Channel-state reports and rate comparisons	Feedback messages and comparisons per interval
Growing quantity	$U \times S$ candidate comparisons	512 for 8 users × 64 subcarriers
Resource problem	Unequal service; weak-user starvation	Per-user rate and Jain index
Root cause	Greedy instantaneous-rate objective	Allocation counts per user

Q7B – Define the Design Objective (2 Marks)

The redesign objective is to guarantee a minimum service level to every admitted user whenever the total system capacity makes that guarantee feasible. After minimum service is satisfied, remaining subcarriers should be allocated using channel-aware selection to improve aggregate throughput.

Define the minimum throughput target as $R_{min} = 256$ kbit/s per active user for this illustrative lecture-hall model. With eight users, the total minimum demand is 2.048 Mbit/s. This target is an explicit design assumption and must be checked against the model's total achievable capacity. If the system cannot meet the target for all users, admission control must identify the infeasible load.

The measurable objectives are: (1) every admitted user has throughput at least R_{min} in feasible test conditions; (2) Jain's fairness index improves compared with pure greedy allocation; and (3) sum throughput should remain as high as possible, with any throughput reduction quantified.

Objective	Measure	Success condition
Minimum service	Minimum per-user throughput	≥ 256 kbit/s when feasible
Fairness	Jain's fairness index	Higher than greedy baseline
Efficiency	Sum throughput	Report trade-off; avoid unnecessary loss
Feasibility	Total capacity vs total minimum demand	Admit only if feasible

Q7C – Revised Strategy

The revised strategy has two stages. Stage 1 is minimum-rate protection. At the start of each scheduling interval, calculate each user's accumulated service over a defined window. Users below the minimum service target receive priority until their deficit is reduced. Stage 2 is channel-aware allocation. Subcarriers not required for the minimum service are assigned to the user with the best proportional-fair utility.

A proportional-fair utility can be defined as instantaneous rate divided by the user's recent average throughput. This rewards users whose current channel is good relative to their own historical service, rather than always selecting the user with the highest absolute rate. The minimum-rate stage ensures that proportional fairness does not leave a user below the explicit service floor when capacity is feasible.

The redesign introduces state: a recent-throughput estimate and a service-deficit counter for each user. It also adds a feasibility check and more scheduler logic. The trade-off is additional state and computation in exchange for a more balanced service distribution.

Feature	Original greedy	Redesigned scheduler
Selection	Highest instantaneous rate	Minimum-rate priority, then proportional-fair utility

History	Not required	Recent average throughput stored
Minimum guarantee	None	Explicit target when feasible
Complexity	Low	Moderate
Risk	Starvation	Possible lower sum rate and extra state

Q7C – Pseudocode for the Redesigned Scheduler

The pseudocode below describes the decision process. It assumes the rate matrix contains the achievable rate for each user on each subcarrier during the current scheduling interval. The minimum number of subcarriers needed by a user depends on the rate of each candidate subcarrier; therefore, the algorithm checks the accumulated assigned rate rather than assigning a fixed number of subcarriers to every user.

Pseudocode:

1. Read current rate matrix R[user, subcarrier].
2. Update each user's recent average throughput.
3. Compute each user's service deficit = max(0, R_min − delivered_rate_in_window).
4. While unassigned subcarriers remain and any feasible user has a deficit, select the user with the largest normalised deficit and assign its best remaining subcarrier.
5. Update that user's predicted delivered rate and deficit.
6. Assign remaining subcarriers to the user/subcarrier pair with the highest proportional-fair utility: R[u,s] / average_rate[u].
7. Record allocation, throughput, fairness and whether any minimum target was infeasible.

The algorithm should use deterministic tie-breaking so that repeated runs with the same input and seed produce the same result. If the model includes a maximum number of subcarriers per user or service class, enforce it explicitly.

- Do not allocate a subcarrier twice
- Remove an assigned subcarrier from the available set
- Use the same channel matrix for original and redesigned methods
- Record the reason for each assignment: minimum-rate protection or proportional-fair stage

Q7C – Flowchart Description

Flowchart sequence: Start → Generate or receive channel-gain matrix → Calculate achievable rate for every user and subcarrier → Update recent average throughput → Calculate each user's service deficit → Are there unassigned subcarriers and a feasible user below the minimum? If yes, choose the user with the greatest normalised deficit, allocate its best available subcarrier, update its predicted rate and repeat the feasibility test. If no, allocate remaining subcarriers using proportional-fair utility. Finally, calculate per-user throughput, sum throughput and Jain's fairness index → Save results → End.

The decision diamond is the feasibility test. The two allocation branches must merge.

Impossible minimum rate: if capacity is insufficient, it records the unmet target and triggers admission control or a lower service profile.

For a hand-drawn or diagram-tool version, use an oval for Start/End, rectangles for calculations and allocation operations, a diamond for the feasibility decision, and arrows to show the repeated allocation loop.

Flowchart symbol	Use
Oval	Start and End
Rectangle	Calculate rates, update averages, allocate subcarrier
Diamond	Any feasible user still below minimum?
Arrow	Control flow and loop
Parallelogram	Input channel matrix and output measurements

Q7D – Prediction Before Implementation (1 Mark)

Before implementation, predict that the redesigned scheduler will increase the minimum per-user throughput and Jain's fairness index compared with pure greedy allocation, provided that total capacity is sufficient to meet the minimum-rate target.

The redesigned method may reduce aggregate throughput because some subcarriers will be assigned to users with weaker instantaneous channels to satisfy minimum service. It will also require additional per-user state and scheduler operations. The prediction is testable by running both methods on identical channel matrices and comparing the same metrics.

A valid prediction must be recorded before examining the redesigned results. If the measured result differs, explain why rather than changing the prediction after the experiment.

Metric	Prediction
Minimum per-user throughput	Increase
Jain's fairness index	Increase
Sum throughput	May decrease modestly
Scheduler state and computation	Increase
Starvation events	Decrease

Q7E – Implement and Validate (2 Marks)

Run the original greedy scheduler and redesigned scheduler on the same 1,000 or more channel realisations. Use identical users, subcarriers, fading distributions, bandwidth, noise assumptions and random seeds. This paired comparison isolates the effect of the allocation rule.

For every realisation, save per-user throughput, sum throughput, Jain's fairness index, minimum throughput, number of users below R_min, and subcarrier allocation counts. Also record the number of rate comparisons and scheduler decisions as supporting complexity evidence.

A suitable validation table is shown below. Fill the values from the executed prototype. The report should include mean and standard deviation across realisations, and a graph comparing the per-user throughput distributions.

Measure	Original greedy	Redesigned	Change
Mean sum throughput	Run output	Run output	Calculate
Mean Jain index	Run output	Run output	Calculate
Mean minimum user rate	Run output	Run output	Calculate
Users below R_min	Run output	Run output	Calculate
Scheduler comparisons	Run output	Run output	Calculate

Q7F – Defend the Final Design (2 Marks)

The final design is successful if the minimum service and fairness objectives improve under feasible load, even if sum throughput falls slightly. A design should not be judged using only aggregate throughput when the stated service requirement applies to every student.

The new costs are additional scheduler state, recent-throughput tracking, more complex decisions and potentially more channel feedback. The algorithm may also need admission control when the number of users or their minimum rates exceed available capacity. If the system is lightly loaded and all users already receive adequate service, the original greedy scheduler may be preferable because it is simpler and can maximise instantaneous throughput.

The conclusion should report the actual magnitude of improvement using the measured values. For example, calculate fairness improvement as J_new − J_old and percentage throughput change as 100 × (T_new − T_old) / T_old. Do not insert hypothetical percentages into the final conclusion.

Question	Defence required
Was prediction supported?	Compare measured fairness and minimum rate
Magnitude of improvement?	Report absolute and percentage change
New costs?	State extra state, feedback and computation
When is original preferable?	Light load or throughput-only objective
What if redesign fails?	Explain feasibility or model assumptions

Appendix A – Reproducible Simulation Protocol

The following protocol should be followed for all questions. Use one project folder with separate scripts for Q1 through Q7, a data folder for CSV outputs, and a results folder for plots. Keep a README file containing the Python version, package versions, command to run each experiment, fixed random seeds and parameter assumptions.

Use a fixed seed for reproducibility, but also run multiple independent seeds when estimating performance variability. A single seed makes the result repeatable; it does not establish that the result is statistically representative. For comparative experiments, use the same generated inputs for all methods being compared.

Each CSV should contain enough columns to reproduce the tables and graphs. Never save only an average if per-user or per-trial observations are available. Raw records allow the evaluator to verify calculations.

- q1_store_forward.csv: availability, update_id, generated_minute, delivered_minute, status
- q2_traffic.csv: dataset, scheme, request_id, arrival, duration, accepted, resource_id, payload_active_time
- q3_cdma.csv: code_length, users, condition, snr_db, power_ratio, timing_offset, bits, errors, ber
- q4_ofdma.csv: trial, strategy, user_id, throughput, allocated_subcarriers, fairness
- q5_sdma.csv: array_size, separation_deg, beamwidth_deg, sidelobe_db, sir_db
- q7_redesign.csv: trial, strategy, user_id, throughput, min_rate_met, sum_rate, fairness

Appendix B – Suggested Python Project Structure

A clean project structure makes the implementation easier to inspect and defend. The scripts should separate input generation, algorithm implementation, evaluation and plotting. Reuse helper functions where possible, but retain a separate entry point for each question so that an evaluator can reproduce a specific result.

Suggested-directory-structure: mobile-computing

```
README.md
requirements.txt
q1_mobile_simulation.py
q2_fdma_tdma.py
q3_cdma.py
q4_ofdma.py
q5_sdma.py
q6_decision_matrix.py
q7_redesign.py
data/
results/
figures/
```

The README should explain the problem, assumptions, execution command, output files and limitations. The GitHub repository should contain no private credentials or personal data. Include a short explanation of how the random seed is set and how each output file is generated.

- Use Python, NumPy, Pandas and Matplotlib for the simulations
- Use fixed random seeds and record package versions
- Keep raw CSV files separate from summary tables

- Name graphs with the question and metric, such as q4_jain_fairness.png

Appendix C – Core Formulas Reference

Q1 expected immediate deliveries: E[D] = Np; expected losses: E[L] = N(1 − p). For a retry-until-success model with independent availability, expected attempts are 1/p. Expected waiting time after an initial failed opportunity, with one-minute retry intervals, is (1 − p)/p minutes.

Q2 nominal FDMA channels = total bandwidth / channel bandwidth. Nominal TDMA traffic slots = number of carriers × slots per carrier. Usable capacity must subtract stated guard, control and synchronisation overheads. Payload utilisation = payload resource-time / available resource-time.

Q3 processing gain = L; processing gain in dB = 10 log10(L). BER = bit errors / total transmitted bits. Normalised code correlation = (1/L)Σa[k]b[k].

Q4 subcarrier spacing Δf = 1/T. Jain's fairness index = (ΣR_i)^2 / (UΣR_i²).

Q5 array factor AF(θ) = Σ exp(jnψ); normalised magnitude = |AF|/N. SIR in dB = 10 log10(P_desired/P_interference).

Q6 weighted score = Σ(weight_j × rating_j), with weights summing to 1. Q7 percentage change = 100 × (new − original) / original.

Question	Formula
Q1	E[delivered] = Np
Q2	FDMA channels = B / channel width
Q3	PG_dB = 10 log10(L)
Q4	Δf = 1/T; Jain = $(\Sigma R)^2/(U\Sigma R^2)$
Q5	AF = Σ exp(jnψ)
Q6	Weighted score = $\Sigma w_i r_i$
Q7	% change = 100(new−old)/old

Appendix D – Graphs to Include in the Final Submission

The assignment asks for tables and graphs. Include graphs that directly answer the inquiry rather than decorative charts. Every graph should have a descriptive title, labelled axes, units, legend where needed and a short interpretation in the report.

For Q1, plot delivered, delayed and lost updates for each availability level and compare mean delivery delay. For Q2, plot utilisation and blocking for FDMA and TDMA across the three traffic datasets. For Q3, plot BER against number of users for each code length and condition, and include a correlation heatmap or table.

For Q4, plot sum throughput by allocation strategy and per-user throughput to expose fairness. For Q5, plot normalised array patterns for N = 2, 4, 8 and 16 and SIR against angular separation. For Q7, compare original and redesigned per-user throughput and fairness across the same channel realisations.

- Use bar charts for method-level comparisons

- Use per-user plots to reveal starvation hidden by averages
- Include the number of trials and random seed in captions

Appendix E – Reference and Standards Guidance

The supplied assignment requires standards parameters such as frame durations, rates and header sizes to be obtained from cited references. The worked assumptions in this report are intentionally labelled as assumptions where the question does not specify a value. Before final submission, add references for any standard-specific parameter used in the code.

Useful primary reference families include 3GPP technical specifications for GSM/GERAN and later radio systems, ITU-R recommendations on radio interfaces and spectrum, and authoritative textbooks on wireless communications, digital communications and antenna arrays. Use the exact document number, version, publication date and relevant section when citing a standard.

For the mathematical foundations, cite a recognised wireless communications textbook for FDMA, TDMA, CDMA, OFDM/OFDMA, link budgets and fading; cite an antenna-array textbook or lecture reference for the ULA array factor and beamwidth. Do not cite a generic blog for a standard-specific timing parameter if a primary specification is available.

- 3GPP specifications: GSM/GERAN architecture, radio interface and procedures
- ITU-R recommendations: radio interface and spectrum terminology
- Wireless communications textbook: multiple access, fading and OFDM
- Antenna-array textbook: array factor, beamwidth and side lobes
- NumPy documentation: random generation, FFT and numerical arrays

Mark Distribution

Question	Assessed Area	Marks
Q1	Mobile communication vs. mobile computing	5
Q2	FDMA and TDMA	6
Q3	CDMA	7
Q4	OFDMA	6
Q5	SDMA	6
Q6	Comparison of SDMA, TDMA, FDMA, CDMA, OFDMA	7
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	13
TOTAL		50

Submission Requirements (Digital)

- Assignment report (PDF) containing the required evidence for Q1–Q7. No handwritten work.
- Source code for all implementations/simulations (GitHub repository link with README and fixed random seeds).
- Raw experimental data (CSV) used to produce your results.
- Tables and graphs used in your analysis.
- Counted quantities (slots, messages, bits, blocked calls, codes, throughput) are primary experimental evidence. Execution time may be reported as supporting evidence but must not be the sole basis for conclusions.
- Standards parameters (frame durations, rates, header sizes, etc.) must be obtained from cited references; state assumptions wherever a value is assumed.

Assessment Verification

You may be asked to analyse any part of your design, code, test-data design, experimental results, predictions, analysis and Design Challenge in a brief verification interpretion/viva. Disclose any AI or reference tools used. You must be able to analyse and defend the evidence as your own work.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook descriptions of mobile systems. It assesses your ability to:

Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate

Your conclusions and design decisions must be supported by your own reasoning and experimental evidence.